Physics 225, Spring 2026: Homework 9 - Solutions
Due 4/9/26

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

1. **Gradient practice.** (40 points) Compute the gradients of the following functions f in Cartesian, cylindrical, and spherical coordinates. For the non-Cartesian coordinate systems, first use the formula for the gradient in terms of the non-Cartesian unit vectors, and then use the conversions between the unit vectors to convert your answer back to Cartesian coordinates. In all cases, you should find the same answer independent of the coordinate system! (You will find that, for all of these examples, the gradient is easy in one coordinate system but a mess in at least one of the others; this illustrates the value of choosing a good set of coordinates for the problem at hand.)

   (a) $f(x,y,z) = x^2 + y^2 + z^2$
   Cartesian: $\nabla f = 2x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 2z\hat{\mathbf{z}}$.
   Cylindrical: $f = \rho^2 + z^2$, $\nabla f = 2\rho\hat{\boldsymbol{\rho}} + 2z\hat{\mathbf{z}}$. Using $\hat{\boldsymbol{\rho}} = \cos\phi\hat{\mathbf{x}} + \sin\phi\hat{\mathbf{y}}$, and $\cos\phi = x/\rho$ and $\sin\phi = y/\rho$, we have $\nabla f = 2\rho\left(\frac{x}{\rho}\hat{\mathbf{x}} + \frac{y}{\rho}\hat{\mathbf{y}}\right) + 2z\hat{\mathbf{z}} = 2x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 2z\hat{\mathbf{z}}$.
   Spherical: $f = r^2$, $\nabla f = 2r\hat{\mathbf{r}}$. Similar to cylindrical coordinates, we use $\hat{\mathbf{r}} = \sin\theta\cos\phi\hat{\mathbf{x}} + \sin\theta\sin\phi\hat{\mathbf{y}} + \cos\theta\hat{\mathbf{z}} = \frac{x\hat{\mathbf{x}}+y\hat{\mathbf{y}}+z\hat{\mathbf{z}}}{r}$, so $\nabla f = 2(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}})$.

   (b) $f(x,y,z) = \sin z$
   Cartesian: $\nabla f = \cos z\hat{\mathbf{z}}$.
   Cylindrical: Since f only depends on z, which is also a coordinate in the cylindrical system, the answer is identical in cylindrical coordinates.
   Spherical: $f = \sin(r\cos\theta)$, $\nabla f = \cos\theta\cos(r\cos\theta)\hat{\mathbf{r}} + \frac{1}{r}(-r\sin\theta)\cos(r\cos\theta)\hat{\boldsymbol{\theta}}$. Substituting $\hat{\mathbf{r}} = \sin\theta\cos\phi\hat{\mathbf{x}} + \sin\theta\sin\phi\hat{\mathbf{y}} + \cos\theta\hat{\mathbf{z}}$ and $\hat{\boldsymbol{\theta}} = \cos\theta\cos\phi\hat{\mathbf{x}} + \cos\theta\sin\phi\hat{\mathbf{y}} - \sin\theta\hat{\mathbf{z}}$, and replacing $r\cos\theta = z$ in the argument of the cosine just for simplicity, we have

$$\begin{aligned}\nabla f &= \cos z\left[\cos\theta(\sin\theta\cos\phi\hat{\mathbf{x}} + \sin\theta\sin\phi\hat{\mathbf{y}} + \cos\theta\hat{\mathbf{z}}) - \sin\theta(\cos\theta\cos\phi\hat{\mathbf{x}} + \cos\theta\sin\phi\hat{\mathbf{y}} - \sin\theta\hat{\mathbf{z}})\right] \\ &= \cos z(\cos^2\theta + \sin^2\theta)\hat{\mathbf{z}} \\ &= \cos z\hat{\mathbf{z}}\end{aligned}$$

   (c) $f(x,y,z) = x + y + z$
   Cartesian: $\nabla f = \hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}$

Cylindrical: $f = \rho\cos\phi + \rho\sin\phi + z$.

$$
\begin{aligned}
\nabla f &= (\cos\phi + \sin\phi)\hat{\boldsymbol{\rho}} + \frac{1}{\rho}(-\rho\sin\phi + \rho\cos\phi)\hat{\boldsymbol{\phi}} + \hat{\mathbf{z}} \\
&= (\cos\phi + \sin\phi)(\cos\phi\hat{\mathbf{x}} + \sin\phi\hat{\mathbf{y}}) + (-\sin\phi + \cos\phi)(-\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}}) + \hat{\mathbf{z}} \\
&= (\cos^2\phi + \sin^2\phi)\hat{\mathbf{x}} + (\cos^2\phi + \sin^2\phi)\hat{\mathbf{y}} + \hat{\mathbf{z}} \\
&= \hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}
\end{aligned}
$$

Spherical: $f = r\sin\theta\cos\phi + r\sin\theta\sin\phi + r\cos\theta$.

$$
\begin{aligned}
\nabla f &= (\sin\theta\cos\phi + \sin\theta\sin\phi + \cos\theta)\hat{\mathbf{r}} + \frac{1}{r}(r\cos\theta\cos\phi + r\cos\theta\sin\phi - \sin\theta)\hat{\boldsymbol{\theta}} \\
&+ \frac{1}{r\sin\theta}(-r\sin\theta\sin\phi + r\sin\theta\cos\phi)\hat{\boldsymbol{\phi}}
\end{aligned}
$$

Substituting the unit vectors and collecting trig functions gives $\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}$.

2. **Divergence practice.** (40 points) Compute the divergences of the following vector fields $\mathbf{v}$ in the given coordinate systems, checking as in problem 1 that you get the same answer (which should just be a scalar function) in all coordinate systems by converting your final answer back to Cartesian coordinates.

   (a) $\mathbf{v} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$: Cartesian, cylindrical, spherical
   Cartesian: $\nabla \cdot v = 1 + 1 + 1 = 3$.
   Cylindrical: $\mathbf{v} = \rho\hat{\boldsymbol{\rho}} + z\hat{\mathbf{z}}$, $\nabla \cdot v = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho^2) + 1 = 2 + 1 = 3$.
   Spherical: $\mathbf{v} = r\hat{\mathbf{r}}$, $\nabla \cdot v = \frac{1}{r^2}\frac{\partial}{\partial r}(r^3) = 3$.

   (b) $\mathbf{v} = \hat{\boldsymbol{\rho}}$: Cartesian, cylindrical
   Cartesian: $\mathbf{v} = \frac{x}{\sqrt{x^2+y^2}}\hat{\mathbf{x}} + \frac{y}{\sqrt{x^2+y^2}}\hat{\mathbf{y}}$.

$$
\begin{aligned}
\nabla \cdot \mathbf{v} &= \left(\frac{1}{\sqrt{x^2+y^2}} - \frac{x^2}{(x^2+y^2)^{3/2}}\right) + \left(\frac{1}{\sqrt{x^2+y^2}} - \frac{y^2}{(x^2+y^2)^{3/2}}\right) \\
&= \frac{2}{\sqrt{x^2+y^2}} - \frac{x^2+y^2}{(x^2+y^2)^{3/2}} = \frac{1}{\sqrt{x^2+y^2}}.
\end{aligned}
$$

   Cylindrical: $\nabla \cdot \mathbf{v} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho) = \frac{1}{\rho} = \frac{1}{\sqrt{x^2+y^2}}$.

   (c) $\mathbf{v} = \hat{\boldsymbol{\theta}}$: Cartesian, spherical

Cartesian: $\mathbf{v} = \frac{xz}{\sqrt{x^2+y^2+z^2}\sqrt{x^2+y^2}}\hat{\mathbf{x}} + \frac{yz}{\sqrt{x^2+y^2+z^2}\sqrt{x^2+y^2}}\hat{\mathbf{y}} - \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}}\hat{\mathbf{z}}$.

$$\begin{aligned}
\nabla\cdot\mathbf{v} &= \left(-\frac{x^2 z}{(x^2+y^2+z^2)^{3/2}\sqrt{x^2+y^2}} - \frac{x^2 z}{\sqrt{x^2+y^2+z^2}(x^2+y^2)^{3/2}} + \frac{z}{\sqrt{x^2+y^2+z^2}\sqrt{x^2+y^2}}\right) \\
&+ \left(-\frac{y^2 z}{(x^2+y^2+z^2)^{3/2}\sqrt{x^2+y^2}} - \frac{y^2 z}{\sqrt{x^2+y^2+z^2}(x^2+y^2)^{3/2}} + \frac{z}{\sqrt{x^2+y^2+z^2}\sqrt{x^2+y^2}}\right) \\
&+ \frac{z\sqrt{x^2+y^2}}{(x^2+y^2+z^2)^{3/2}} \\
&= \frac{z^3}{\sqrt{x^2+y^2}(x^2+y^2+z^2)^{3/2}} + \frac{z\sqrt{x^2+y^2}}{(x^2+y^2+z^2)^{3/2}} \\
&= z\left(\frac{x^2+y^2+z^2}{\sqrt{x^2+y^2}(x^2+y^2+z^2)^{3/2}}\right) \\
&= \frac{z}{\sqrt{x^2+y^2+z^2}\sqrt{x^2+y^2}}.
\end{aligned}$$

Spherical:

$$\begin{aligned}
\nabla\cdot\mathbf{v} &= \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta) \\
&= \frac{\cot\theta}{r} \\
&= \frac{1}{\sqrt{x^2+y^2+z^2}}\left(\frac{z}{\sqrt{x^2+y^2}}\right)
\end{aligned}$$

3. **Rotational invariance of the divergence.** (20 points) In index notation with the summation convention, the divergence of a vector field v^i can be written as $\partial_i v^i$. Show that the divergence is *invariant* under a rotation of the coordinate system. To do this, use the coordinate transformation defined in Part 3 of Discussion 9, $x^i = R^i{}_j x'^j$, which can be inverted to give $x'^i = (R^{-1})^i{}_j x^j$. In other words, a vector with an upper index transforms with a factor of R^{-1}, while the gradient with a lower index transforms with a factor of R (as shown in Discussion 9). Showing the invariance of the divergence is now just an exercise in careful index placement: make sure not to repeat any dummy indices in more than one summation!

From Discussion 9, $\partial'_i = R^k{}_i \partial_k$. So $\partial'_i v'^i = (R^k{}_i \partial_k)(R^{-1})^i{}_j v^j$. The components of the coordinate transformation matrix R are constants, so we can move the derivative past them and get

$$\partial'_i v'^i = R^k{}_i (R^{-1})^i{}_j \partial_k v^j = \delta^k{}_j \partial_k v^j = \partial_j v^j,$$

which is the same as $\partial_i v^i$ because the dummy index can have any name.